

- [Charts: visualizing data](#)
- [Getting started with BigQuery](#)

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the [full course website](#) for more.

- [Intro to Pandas DataFrame](#)
- [Intro to RAPIDS cuDF to accelerate pandas](#)
- [Getting Started with cuML's accelerator mode](#)

Using Accelerated Hardware

- [Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API](#)
- [Train a Vision Transformer \(ViT\) for image classification with JAX](#)
- [Text classification with a transformer language model using JAX](#)

▼ Featured examples

- [Train a miniGPT language model with JAX AI Stack](#)
- [LoRA/QLoRA finetuning for LLM using Tunix](#)
- [Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA](#)
- [Loading Hugging Face Transformers Checkpoints](#)
- [8-bit Integer Quantization in Keras](#)
- [Float8 training and inference with a simple Transformer model](#)
- [Pretraining a Transformer from scratch with KerasHub](#)
- [Simple MNIST convnet](#)
- [Image classification from scratch using Keras 3](#)
- [Image Classification with KerasHub](#)

```
import numpy as np
import matplotlib.pyplot as plt

house_data = np.array([
    [3, 1500, 250000],
    [5, 2800, 450000],
    [4, 2000, 320000],
    [6, 3500, 600000],
    [5, 3000, 500000]
])

filtered_houses = house_data[house_data[:, 0] > 4]

average_sale_price = np.mean(filtered_houses[:, 2])

print("Houses with more than 4 bedrooms:")
print(filtered_houses)

print("\nAverage Sale Price:", average_sale_price)

prices = filtered_houses[:, 2]
labels = [f"House {i+1}" for i in range(len(prices))]

plt.figure(figsize=(8, 5))

bars = plt.bar(labels, prices)
plt.axhline(average_sale_price, color='red', linestyle='--', label='Average Price')

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 5000,
        f"{height:.0f}",
        ha="center"
    )
```

```
plt.title("Sale Prices of Houses with more than 4 bedrooms ")
plt.xlabel("Houses")
plt.ylabel("Sale Price")
plt.legend()
plt.grid(axis="y", linestyle="--", alpha=0.7)

plt.show()
```

Houses with more than 4 bedrooms:

```
[[ 5 2800 450000]
 [ 6 3500 600000]
 [ 5 3000 500000]]
```

Average Sale Price: 516666.6666666667

